

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First / Second Semester 20-20****

Mid-Semester Test
(EC-2 Regular/ EC-2 Make-up)

Course No. :
 Course Title : Conversational AI
 Nature of Exam : Closed Book (As per Course Handout)
 Weightage : 30% (As per Course Handout)
 Duration : 2 Hours
 Date of Exam :

No. of Pages	= 2
No. of Questions	= 5

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1	Given the scenario: “A user asks a conversational agent to track the delivery status of an online order.” Explain the scenario how a modern conversational agent processes this request using the 7-step pipeline: Request-Routing-Reasoning-Tool Invocation-Memory-Safety-Response.	(7 Marks)																
	i. Request The agent receives the user query, identifies the intent as an order tracking request, extracts key entities such as the order ID, and validates the input. ii. Routing The request is classified as an e-commerce service query and routed to the order management or logistics workflow. iii. Reasoning The LLM plans the required steps, including verifying the order ID and deciding which backend systems must be queried to obtain delivery information. iv. Tool Invocation The agent invokes external order management and logistics APIs to fetch the current shipment status and expected delivery time. v. Memory The agent stores the interaction context, such as recently tracked orders, to improve continuity in future conversations. vi. Safety Privacy and security checks are applied to ensure that order details are shared only with the authenticated user and sensitive information is protected. vii. Response The agent generates a natural language response and informs the user of the current delivery status in a clear and concise manner.																	
Q2	Compare Encoder-Only, Decoder-Only, and Encoder–Decoder Transformer architectures. Explain which architecture is most suitable for text classification, machine translation, and code synthesis, with justification based on their context-processing mechanisms.	(6 Marks)																
	<table><tr><th>Architecture</th><th>Context Processing</th><th>Best Task</th><th>Reason</th></tr><tr><td>Encoder-Only</td><td>Bidirectional self-attention</td><td>Text Classification</td><td>Needs full-context understanding</td></tr><tr><td>Decoder-Only</td><td>Masked self-attention (causal)</td><td>Code Synthesis</td><td>Requires sequential, step-wise generation</td></tr><tr><td>Encoder–Decoder</td><td>Bidirectional + cross-attention</td><td>Machine Translation</td><td>Needs strong input–output alignment</td></tr></table>	Architecture	Context Processing	Best Task	Reason	Encoder-Only	Bidirectional self-attention	Text Classification	Needs full-context understanding	Decoder-Only	Masked self-attention (causal)	Code Synthesis	Requires sequential, step-wise generation	Encoder–Decoder	Bidirectional + cross-attention	Machine Translation	Needs strong input–output alignment	2 marks each
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Q3	Explain (with flow diagrams) the three steps (SFT, Reward model and PPO) in training the InstructGPT model	(6 Marks)								
	<table><tr><th>Step</th><th>Explanation</th></tr><tr><td>1.Supervised Fine Tuning</td><td>Base LLM trained with human demonstrations</td></tr><tr><td>2.Reward Model Training</td><td>Reward model trained with human preference ranking</td></tr><tr><td>4. RLHF/PPO</td><td>Optimize policy using Reward feedback</td></tr></table> Step 1 Collect demonstration data, and train a supervised policy. A prompt is sampled from our prompt dataset. A labeler demonstrates the desired output behavior. This data is used to fine-tune GPT-3 with supervised learning. Step 2 Collect comparison data, and train a reward model. A prompt and several model outputs are sampled. A labeler ranks the outputs from best to worst. This data is used to train our reward model. Step 3 Optimize a policy against the reward model using reinforcement learning. A new prompt is sampled from the dataset. The policy generates an output. The reward model calculates a reward for the output. The reward is used to update the policy using PPO.	Step	Explanation	1.Supervised Fine Tuning	Base LLM trained with human demonstrations	2.Reward Model Training	Reward model trained with human preference ranking	4. RLHF/PPO	Optimize policy using Reward feedback	Explanation: 3 marks and diagram:3 marks
Step	Explanation									
1.Supervised Fine Tuning	Base LLM trained with human demonstrations									
2.Reward Model Training	Reward model trained with human preference ranking									
4. RLHF/PPO	Optimize policy using Reward feedback									
Q4	Why is free-form text output unsuitable for automation systems? What approach can be used to overcome this limitation? Why it fails for automation: Software cannot read sentences. If you try to feed that paragraph into any app or a database, the code breaks because it is looking for specific numbers, not a story. It is unpredictable and inconsistent. (1 mark) No type safety, Ambiguity, requires complex regex parsing, breaks downstream API integrations and automation pipelines (2 marks) Solution: Structured output (1 mark) , example: JSON Schema (1 mark)	(5 Marks)								
Q5	The training objective function of direct preference optimization is $\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w x)}{\pi_{\text{ref}}(y_w x)} - \beta \log \frac{\pi_{\theta}(y_l x)}{\pi_{\text{ref}}(y_l x)} \right) \right]$ a) Explain the main objective of this equation. (2 marks) b) A code review assistant is trained using DPO to prefer secure and efficient code. For a given prompt x, two responses are evaluated: - Secure and optimized code (preferred) - Insecure or inefficient code (rejected) The model and reference probabilities are:	(2+4=6 Marks)								

	<table> <tr> <th>Model</th><th>Reference model (π_{ref})</th><th>Current model (π_{θ})</th></tr> <tr> <td>P(Secure and optimized code x)</td><td>0.50</td><td>0.25</td></tr> <tr> <td>P(Insecure or inefficient code x)</td><td>0.20</td><td>0.40</td></tr> </table>	Model	Reference model (π_{ref})	Current model (π_{θ})	P (Secure and optimized code x)	0.50	0.25	P (Insecure or inefficient code x)	0.20	0.40	
Model	Reference model (π_{ref})	Current model (π_{θ})									
P (Secure and optimized code x)	0.50	0.25									
P (Insecure or inefficient code x)	0.20	0.40									
	Given $\beta=0.1$, compute the DPO loss. (4 MARKS)										
Q5 Sol	a) Maximize the probability of the winning answer and minimize the probability of losing answer. (2 marks) b) y_w: Secure and optimized code (preferred) y_l: Insecure or inefficient code (rejected) Preferred response = $0.5/0.25 = 2$ (0.5 mark) Rejected response = $0.2/0.4 = 0.5$ (0.5 mark) Log ratios = $\log(2) = 0.693$ and $\log(0.5) = -0.693$ (0.5 mark) Difference of log ratios = $0.693 - (-0.693) = 1.386$ (0.5 mark) $\beta \cdot (\text{Difference of log ratios}) = 0.1 \cdot 1.386 = 0.1386$ (0.5 mark) Sigmoid (0.1386) = 0.535 (0.5 mark) Final DPO loss = $-\log(0.535) = 0.63$ (1 mark)										
